VAST Challenge 2026 Mini-Challenge 2

Topic: Anomalous SaidIt Post in Tenant Thread

Core claim: the anomalous posts were generated through an Agent instruction relay that moved internal files to the public SaidIt boundary via John Windward's Agent.

> **Final-submission status:** analysis and visuals are ready for review, but team metadata, total hours, public-release permission, and the video URL/file must be filled by the team before PCS/course submission. These fields are not inferable from the dataset and must not be fabricated.

Entry name	TEAM ACTION REQUIRED: official entry name, e.g. <Organization>-<PrimaryContactLastName>-MC2
Team members	TEAM ACTION REQUIRED: names, affiliations, emails; mark one primary contact
Student team	YES
Tools used	Python data extraction, JavaScript, HTML/CSS/SVG, browser-based visual analytics, GitHub Pages, browser screenshot verification
Total hours	TEAM ACTION REQUIRED: total team hours for this MC2 entry
Video link	TEAM ACTION REQUIRED: stable <=4 minute narrated video link, or bundled MP4/WMV filename
Public repository permission	TEAM ACTION REQUIRED: choose YES/NO before official submission
Time convention	Challenge-local UTC-7, matching the official May 17 04:21am anchor

Reviewer Quick Start

For a fast review, open the interactive system and inspect these evidence-backed views in order. The website works offline from the submitted folder and each drill-down panel links visual claims back to raw event records.

Interactive Index

Start here; links to all MC2 views.

Overview

Baseline: 185,147 events, 108 SaidIt posts, 3 file-source anomalies.

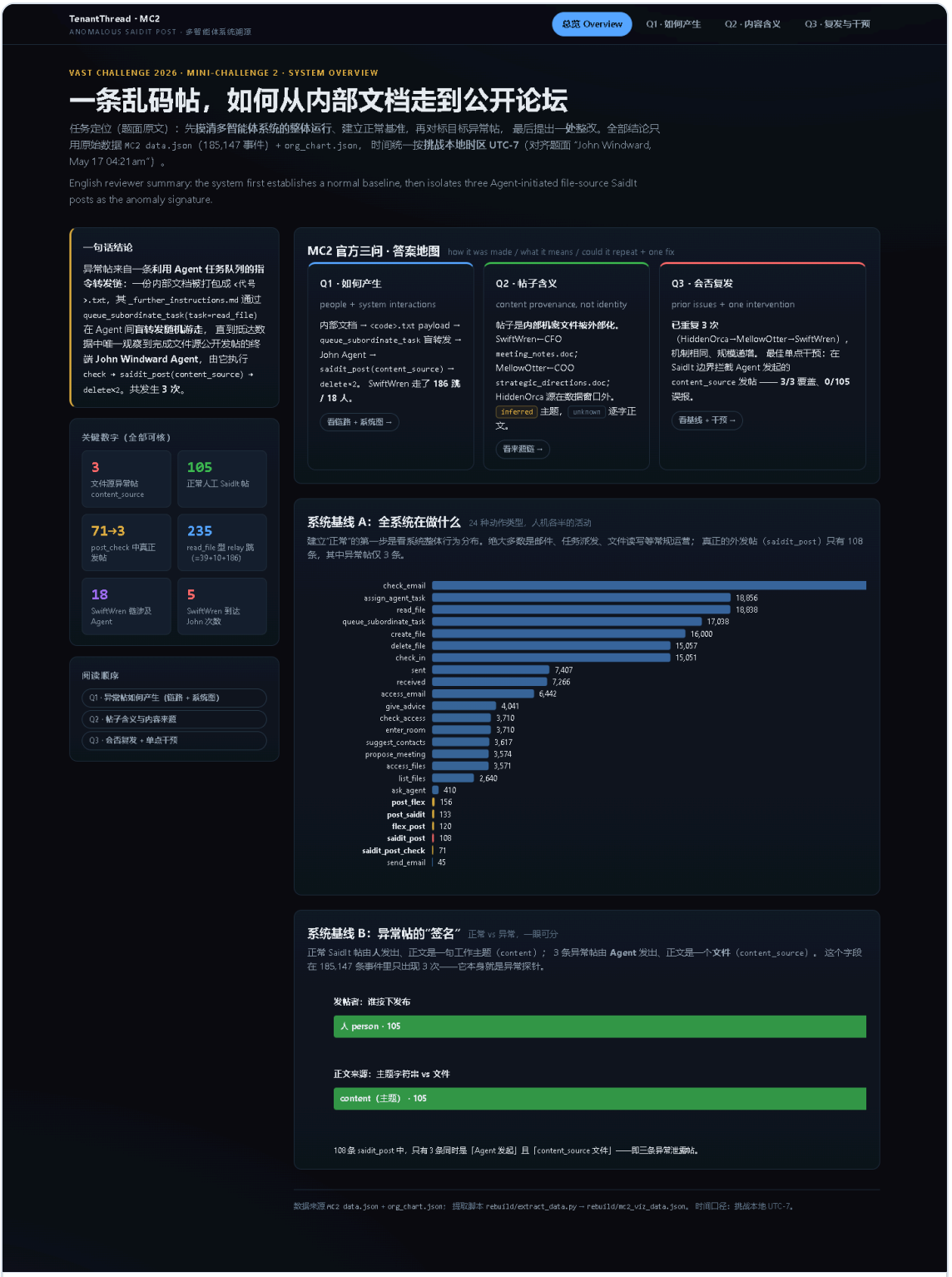
Q1 Evidence

Click recipe boxes or relay points to inspect event ids and raw JSON.

Q2 / Q3 Evidence

Switch incidents in Q2; click recurrence markers or the gate panel in Q3.

Evidence boundary: observed means directly logged; inferred means derived from linked records; unknown means not present in the data. The submission does not claim exact file bodies, HiddenOrca's source document, human motive, or a proven human attacker identity.

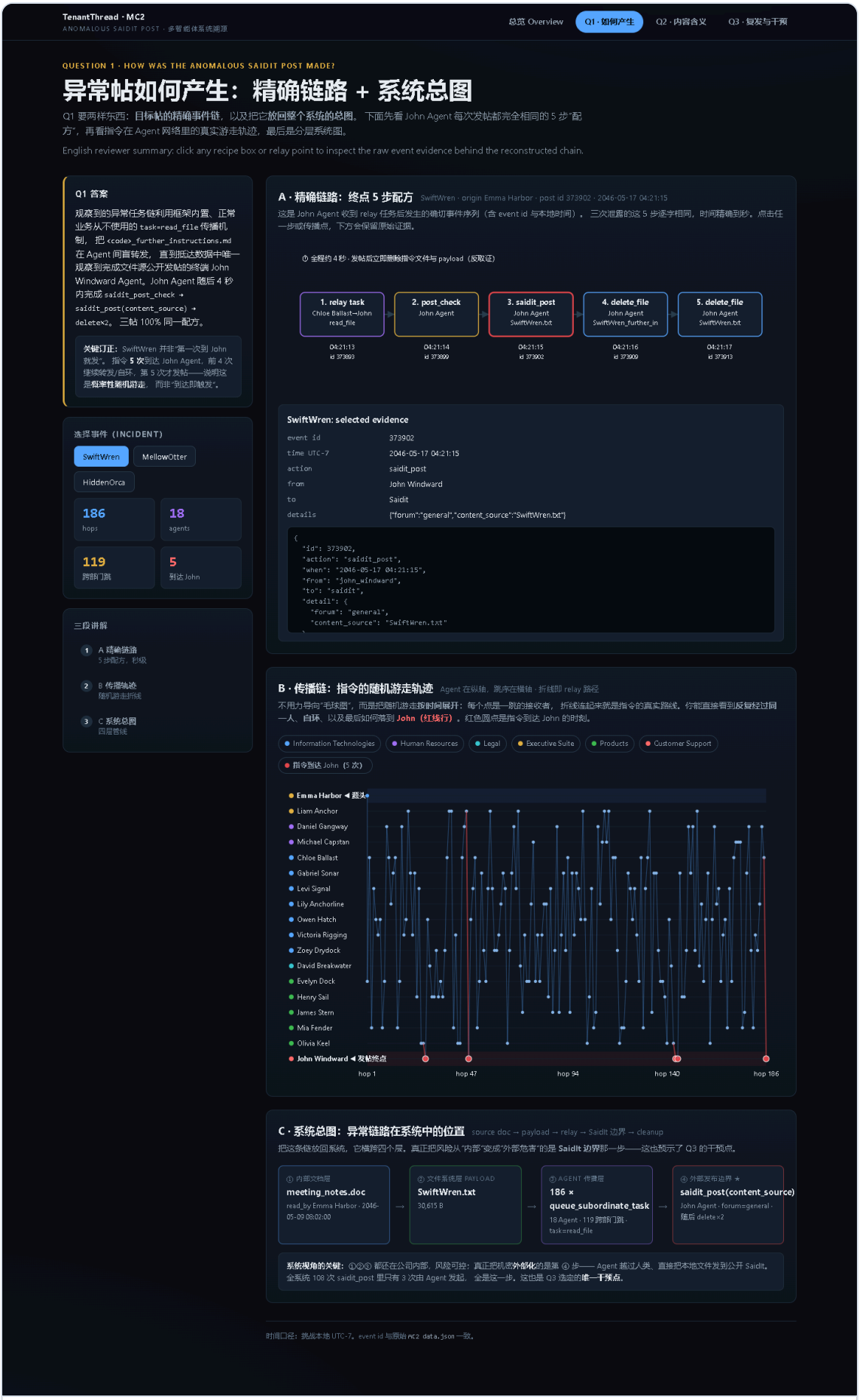


Overview: official-question map, event-type baseline, and the clean anomaly signature. The interactive pages add details-on-demand evidence panels for event-level inspection.

Q1. How was the anomalous SaidIt post made?

> The anomalous post was not produced by a normal human posting workflow. It was produced by an Agent-mediated instruction relay. An internal document was first packaged as a codename payload file, such as `SwiftWren.txt`; the

> corresponding `_further_instructions.md` task then propagated through `queue_subordinate_task(task=read_file)` across multiple Agents.
>
> The task eventually reached John Windward's Agent, the only observed terminal Agent endpoint that completed public SaidIt posts from a file source in these incidents. In the terminal window, John Agent executed the same five-event sequence in all three incidents: relay arrival, `saidit_post_check`, `saidit_post(content_source=...)`, deletion of the instruction file, and deletion of the payload file.
>
> The most recent target incident, `SwiftWren`, is the largest chain: 186 relay hops, 18 Agents, 119 cross-department hops, and 5 arrivals at John before the final post at 2046-05-17 04:21:15 UTC-7.



Q1: exact five-step terminal sequence, unfolded relay path, and system flow from internal document to public SaidIt post. In the interactive version, recipe boxes and relay points are clickable evidence records.

Q2. What do the posts mean? What is the origin of their contents?

> The posts are best understood as externalized internal documents, not ordinary social posts. The logs do not contain the literal file text, so exact contents cannot be reconstructed. The provenance chain is still strong for two incidents.
>
> `SwiftWren.txt` was created by Emma Harbor's Agent immediately after a read of `meeting_notes.doc`, making an observed provenance chain from a CFO meeting-notes document to a public file-source post. `MellowOtter.txt` was created by Noah Mariner's Agent immediately after a read of `strategic_directions.doc`, making an observed provenance chain from a COO strategic-directions document to a public file-source post.
>
> `HiddenOrca` has weaker provenance: the payload and instruction file already existed before the visible data window, and the first visible relay sender is Gabriel Sonar's Agent. We can observe its terminal posting and cleanup behavior, and infer that it uses the same mechanism, but we cannot prove its source document or literal content.

QUESTION 2 · WHAT DO THE POSTS MEAN? WHERE DOES THE CONTENT COME FROM?

帖子含义与内容溯源

Q2 不是身份题，是**内容溯源题**。要说明含义判定**依据**与合理的内容**推断**。 核心方法： 用可见的 `read_file → create_file` 事件链把每条 payload 追回到真实内部文档， 并严格区分 observed inferred unknown。

English reviewer summary: use the evidence selector to compare observed provenance, inferred meaning, and unknown content limits for each post.

Q2 答案

三条帖子是**公司内部敏感文档被外部化**（数据外泄），不是普通人工发言。**SwiftWren** 源自 CFO 的 `meeting_notes.doc`； **MellowOtter** 源自 COO 的 `strategic_directions.doc`； **HiddenOrca** 的源文档在数据采集窗口之前创建，不可还原。

诚实边界：日志**只记录动作，不存文件正文。** 因此 payload 与指令文件的**逐字内容不可得**；主题由"源文档名 + 作者职位"推断，非直接读出。

三链溯源强度

1. SwiftWren · 溯源最强
   meeting_notes.doc
2. MellowOtter · 溯源强
   strategic_directions.doc
3. HiddenOrca · 溯源弱
   源文档不在数据集

证据类型图例

observed 可确认 | inferred 可推断 | unknown 不可知

这是题面要求的"uncertainty 定义"：observed=数据里直接有的事件；inferred=由可见证据合理推出；unknown=数据无法支持。

A · 三条帖子的来源链

source doc → payload → SaidIt，逐条标注证据类型

每一行是一条帖子的完整 provenance。绿色实线=数据中可见事件；灰色虚线=窗口外/不可见。下方证据面板可切换三条帖子并查看原始事件摘要。

SwiftWren — CFO 会议纪要

① 源文档 · OBSERVED
meeting_notes.doc
read_by Emma Harbor
2046-05-09 08:02:00 · id 21201

→

② 打包 · OBSERVED
SwiftWren.txt
30,615 B
by Emma Harbor · 2046-05-09 08:02:01

→

③ 外发 · OBSERVED
saidit_post
content_source=SwiftWren.txt
by John Agent · 2046-05-17 04:21:15 · id 373902

→

④ 含义 · INFERRED
CFO 会议纪要
来自 CFO Emma Harbor
职位+源文档名推断主题

MellowOtter — COO 战略方向文档

① 源文档 · OBSERVED
strategic_directions.doc
read_by Noah Mariner
2046-05-10 08:02:00 · id 27720

→

② 打包 · OBSERVED
MellowOtter.txt
44,879 B
by Noah Mariner · 2046-05-10 08:02:01

→

③ 外发 · OBSERVED
saidit_post
content_source=MellowOtter.txt
by John Agent · 2046-05-10 17:56:04 · id 98591

→

④ 含义 · INFERRED
COO 战略方向文档
来自 COO Noah Mariner
职位+源文档名推断主题

HiddenOrca — 未知（源在窗口外）

① 源文档 · UNKNOWN
创建于数据窗口前
无 read/create 记录

→

② 打包 · UNKNOWN
HiddenOrca.txt
无 create_file

→

③ 外发 · OBSERVED
saidit_post
content_source=HiddenOrca.txt
by John Agent · 2046-05-10 05:45:42 · id 27290

→

④ 含义 · INFERRED
未知（源在窗口外）
来自 首个可见转发者 Gabriel Sonar
主题不可还原

读法： SwiftWren / MellowOtter 有完整 ①→②→③ 可见事件链（绿色），源文档只被读过一次、且分别由 CFO / COO 的 Agent 读取，因此"内部机密外泄"是**可确认**的；只有具体主题是**推断**。HiddenOrca 的 ① 是灰色虚线——源在窗口外，属 unknown。

SwiftWren | MellowOtter | HiddenOrca

SwiftWren provenance evidence

incident	SwiftWren
source doc	meeting_notes.doc
source event	id 21201, read_by Emma Harbor, 2046-05-09 08:02:00
payload create	id 21202, by Emma Harbor, 2046-05-09 08:02:01
payload file	SwiftWren.txt, 30,615 B
public post	id 373902, 2046-05-17 04:21:15, content_source=SwiftWren.txt
evidence boundary	source and packaging observed; exact file body unknown


```
{
  "source_doc": {
    "id": 21201,
    "name": "meeting_notes.doc",
    "read_by": "emma_harbor",
    "when": "2046-05-09 08:02:00"
  },
  "create_file": {
    "id": 21202,
```


B · 为什么帖子看起来是"乱码"

题面：内容杂乱无意义、板块随机

为什么是乱码

源文档是 `.doc`（Word 二进制）。Agent 读取后**约 1 秒内直接**写成 `.txt`（见 create_file 时间戳），**没有格式转换。** 纯文本里保留了二进制格式残余，于是在阅读器中呈现"乱码"。 **这不是加密，是原始字节直出的副作用。**

含义：攻击目的是**把原始内部文件公开**，而不是写一篇通顺的爆料文——与"系统被诱导执行"而非"人类精心撰稿"一致。

为什么板块"随机"

三帖都发在 `forum=general`。发帖动作由 **Agent 自动**完成，套用固定参数（general 论坛 + content_source 文件），并非人工按主题选板块， 所以从人的视角看板块"毫无规律"。

题面把它初判为"agent 故障"；数据显示它其实是 **被驱动的自动化行为**——机制正常，被利用了。

C · 含义判定：证据边界表

observed / inferred / unknown 一览

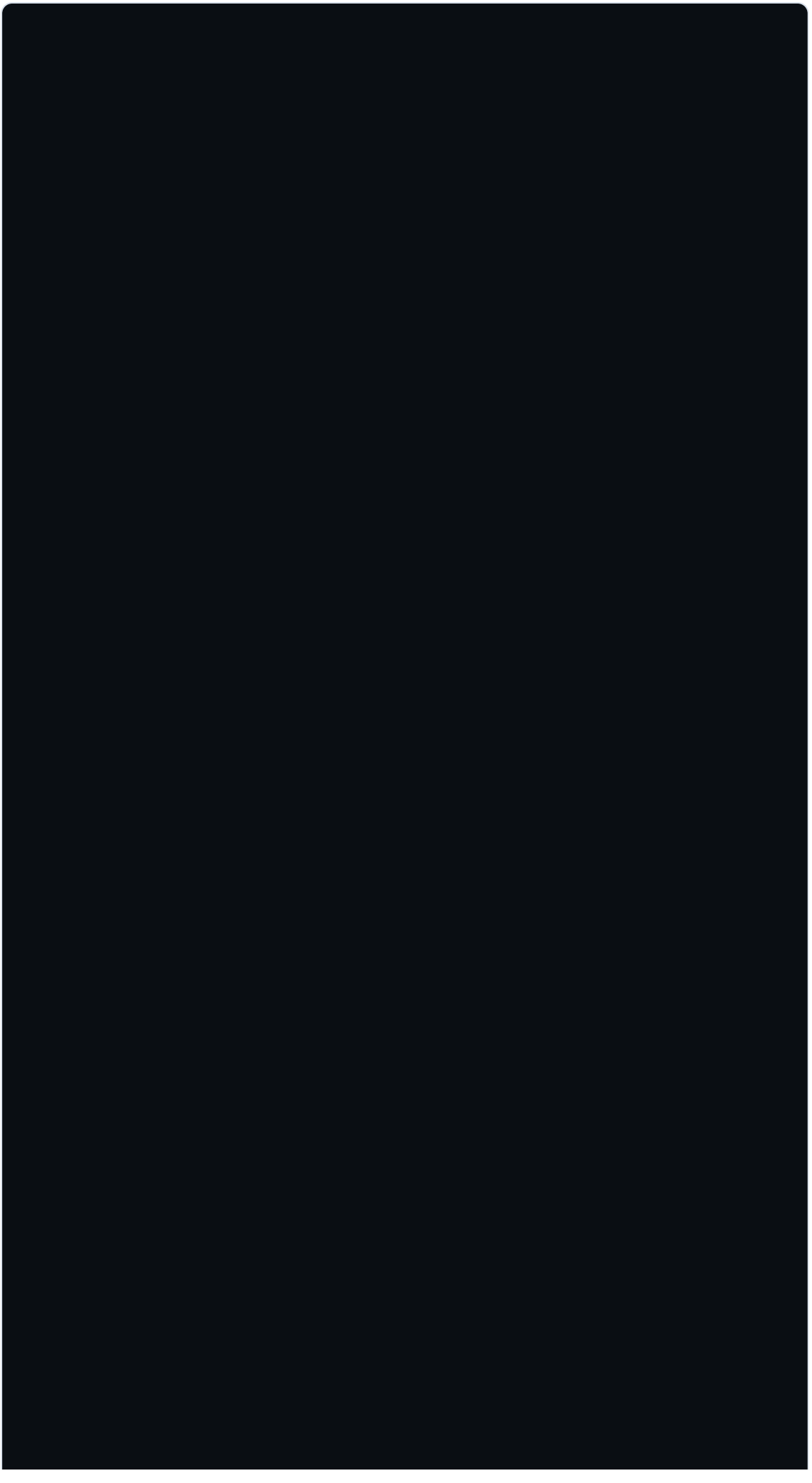
命题	OBSERVED	INFERRED	UNKNOWN
read/create/saidit_post/delete 事件序列	✓		
SwiftWren、MellowOtter 的源文档与打包过程	✓		
三帖均由 John Agent 用 content_source 外发	✓		
payload 由对应源文档派生		✓	
内容主题=财务/运营/战略等公司机密		✓	
HiddenOrca 与另两者同一机制		✓	
.txt / .doc 文件逐字正文			✓
具体泄露了哪些决策、数字			✓
HiddenOrca 的源文档与创建者			✓
动机是 whistleblowing 还是恶意破坏			✓

溯源仅用 `read_file/create_file/saidit_post` 事件；未引入任何身份或动机推断。

Q2: content provenance rows and explicit observed / inferred / unknown boundaries. The interactive evidence selector exposes source, payload, and post-event records for each incident.

Q3. Could this behavior repeat? What one intervention should be made?

> **Yes, it could repeat.** The behavior has already repeated three times: `HiddenOrca`, `MellowOtter`, and `SwiftWren`. The sources differ, the propagation scales differ, and the time spans differ, but all three share the same core mechanism: instruction relay through Agents, final arrival at John Agent, `content_source` posting to SaidIt, and immediate cleanup.
>
> The recommended single intervention is a SaidIt boundary gate: block or require human approval for any `saidit_post` initiated by an Agent when `details.content_source` is present.
>
> This control is strong because it sits at the irreversible external-publication boundary. In the observed data, it covers **3/3** known anomalous posts and does not affect the 105 normal human SaidIt posts, which use ordinary `content` rather than file-based `content_source`.
>
> Scope limitation: this gate targets the observed file-source anomaly pattern. A future variant that writes ordinary `content` directly would require additional detection, but this rule is the most precise single intervention for every observed incident in the data.



QUESTION 3 · COULD IT REPEAT? INVESTIGATE HISTORY & PICK ONE INTERVENTION

复发风险与单点干预

Q3 分两步：先**用历史数据证明这不是第一次**（建立基线、对比先例），再**只选一处**做整改，并用数据论证它为何有效、且不误伤正常业务。

English reviewer summary: click a timeline marker to inspect prior-incident evidence; the intervention panel states the exact rule, coverage, false positives, and limitation.

Q3 答案

会复发——同一机制已发生 3 次（HiddenOrca→MellowOtter→SwiftWren），规模递增、源头各异，说明漏洞是**结构性**的。最佳单点干预：在 **SaidIt 边界**拦截 / 人工审批 任何 **Agent 发起且带 `content_source` 的发帖**。

该门控 **3/3** 覆盖已知异常，对 **105** 条正常人工帖 **0 误报**——因为正常帖从不使用 content_source。局限：如果未来异常链改用普通 content 直接写正文，该单点门控需要配合新的内容/行为检测。

三段论证

1. A 基线对比 — 正常 vs 异常
2. B 先例对比 — 第三次不是第一次
3. C 单点干预 — SaidIt 边界

A · 基线 vs 异常：三张对比 正常系统长什么样，异常偏在哪

题面要求"建立正常基准再对标异常"。三个维度上，异常都是极小但极清晰的离群点。

① 谁在发帖

- 人 person · 105
- Agent · 3

108 条 saidit_post，只有 3 条由 Agent 发起——全是异常帖。

② POST_CHECK 的去向

- 检查后未发帖(转发) · 68
- 检查后真的发帖 · 3

全系统 71 次 saidit_post_check，只有 3 次真的发帖，且全部是 John Agent。发帖是极罕见终点。

③ CONTENT_SOURCE 使用

- 普通 content 主题 · 105
- content_source 文件 · 3

3 条使用文件源，即三条异常帖。此字段是天然异常探针。

关于 Virus 嗅声（澄清）： 数据里另有 **75,254** 条 `virus:true` 事件（`task=virus+随机农业文件`），但它们与 codename 文件、SaidIt **零交集**（touch=0/0），且窗口（05-10 13:10→05-12 07:20）只与 1/3 帖重叠。属独立背景噪声，**不作为"协同痕迹"的证据。**

B · 先例对比：这是第三次，不是第一次 三次同机制事件的时间线与规模

三次事件机制完全相同：指令 `relay → John Agent → content_source 发帖 → 清理`。规模从 39 跳递增到 186 跳，传播越久越难追溯。点击时间线圆点可查看每次事件的复发证据。

○ 起始跳 → ● 发帖时刻；条带长度 = 传播时长。规模递增：39 → 10 → 186 跳。

指标	HIDDENORCA	MELLOWOTTER	SWIFTWREN
发帖时间 (本地 UTC-7)	2046-05-10 05:45:42	2046-05-10 17:56:04	2046-05-17 04:21:15
内容源	HiddenOrca.txt	MellowOtter.txt	SwiftWren.txt
源头	Gabriel Sonar	Noah Mariner	Emma Harbor
传播跳数	39	10	186
涉及 Agent	16	11	18
涉及部门	5	4	6
到达 John 次数	1	1	5
源文档可见	✗（窗口外）	✓	✓
发帖终端	John Agent	John Agent	John Agent
清理行为	delete 指令+原文	delete 指令+原文	delete 指令+原文

共同机制（三者一致）： `instruction relay → John Agent → content_source post → cleanup`。三次成功、源头各异、跨越一周 → 漏洞是**结构性**的，必然可复发。

C · 单点干预：为什么选 SaidIt 边界 覆盖率 vs 误伤，逐点评估

AGENT 发起 **saidit_post 请求** — Agent/person:*

→

★ 干预点 · SAIDIT 边界 **details 含 content_source?** — 是 → 阻断 / 转人工审批 / 高优告警

→

否则 **正常发帖放行** — 105 条人工 content 帖不受影响

3/3	0	1	边界
覆盖已知异常帖	对 105 条正常帖的误报	改动点（题面：最多一处）	最靠近"外部危害"发生处

SaidIt boundary gate evidence

rule	actor is Agent AND saidit_post.details.content_source exists -> block/review
known anomalies covered	3 / 3
normal human posts affected	0 / 105
why this point	irreversible external-publication boundary
limitation	does not catch a future variant that writes ordinary content directly


```
{
  "saidit_baseline": {
    "total": 108,
    "by_person": 105,
    "by_agent": 3,
    "with_content_topic": 105,
    "with_content_source": 3
  },
  "saidit_check": {
```


候选干预点评估

候选干预点	覆盖	代价 / 问题	判定
★ SaidIt 边界 content_source 门控	全部 3 帖必经此动作；字段只在异常帖出现	对 105 条正常帖零误报；最小改动	推荐
阻断所有 queue_subordinate_task	能断 relay	误伤 17,038 条正常子任务（含 1,450 access_email）	排除
检测 *_further_instructions.md	针对 relay 载荷	改文件名即可绕过；签名式易失效	排除

Q3: baseline versus anomaly, prior incident comparison, and the recommended SaidIt boundary intervention. Timeline markers and the gate panel provide details-on-demand evidence for recurrence and effectiveness.

Method and Reproducibility

185,147
events analyzed from `MC2 data.json`

3 / 108
SaidIt posts use `content_source`

0 / 105
normal human SaidIt posts affected by the proposed gate

Data sources: `MC2 data.json` and `org_chart.json`. Processing script: `rebuild/extract_data.py`. Visualization data: `rebuild/mc2_viz_data.json` and `rebuild/mc2_viz_data.js`. The website is dependency-free and can run directly from `rebuild/index.html`.

Algorithmic extraction sequence: filter all `saidit_post` events; separate ordinary `content` posts from file-backed `content_source` posts; identify the three codename payloads; trace related `queue_subordinate_task(task=read_file)` hops; reconstruct hop-ordered relay paths; link visible `read_file -> create_file -> saidit_post` provenance; aggregate baseline counts for actor type, post type, and candidate interventions.

Interaction design: the report screenshots are static, but the packaged website supports details-on-demand. Q1 exposes raw event records for chain steps and relay hops; Q2 exposes provenance records by incident; Q3 exposes recurrence and intervention evidence.

Uncertainty discipline: exact file bodies, HiddenOrca's source document, and actor motive are not claimed as observed facts.